

Graph Entropy

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- **Discrete, Stationary, and Memoryless Source:**

The foundational model relies on an information source that emits symbols independently and with a constant probability distribution over time.

- **Partial Indistinguishability of Symbols:**

Unlike classical models, the alphabet symbols (letters) emitted by the source are not always completely discernible from one another.

- **Distinguishability of Strings:**

Two sequences (strings) of symbols are considered distinguishable if and only if they are distinguishable at least at one coordinate.

- **High-Probability Encoding:**

Proper encoding is only required for “most” strings, allowing indistinguishable strings to share codewords within a strictly bounded, asymptotically vanishing error probability (λ).

- **Encoding Performance and Compression Rate:**

The performance of an encoding is measured by the rate $\frac{\log M}{n}$ (where M is the number of distinct codewords and n is the string length), where a lower rate indicates better compression performance..

- **Intuitive Meaning of Graph Entropy:**

$H(G, P)$ represents the absolute minimum average number of bits needed per symbol to optimally encode a message generated by this specific source.

Presentation Overview

- ① Definitions
- ② Main Theorem - Equivalence of Graph Entropy Definitions
 - Key Lemmas
 - Chromatic Limit and Information Theoretic Definitions
 - Information Theoretic and Geometric Definitions
- ③ Maximum Graph Entropy and Fractional Chromatic Number
- ④ References

Definition 1 (Convex Hull)

For a subset $S \subseteq \mathbb{R}^n$, the *convex hull* of S , denoted by $\text{conv}(S)$, is the smallest convex set containing S . Equivalently:

$$\text{conv}(S) = \left\{ \sum_{i=1}^k \lambda_i x_i \mid k \in \mathbb{N}, x_i \in S, \lambda_i \geq 0, \sum_{i=1}^k \lambda_i = 1 \right\}.$$

Definition 2 (Packing Polytope)

The *vertex packing polytope* $VP(G)$ of a graph G is the convex hull of the characteristic vectors of maximal independent sets of G .

Definition 3 (Convex Corner)

A set $\mathcal{A} \subseteq \mathbb{R}_+^n$ is called a *convex corner* if it is closed, convex, has a non-empty interior, and satisfies the property that if $0 \leq a'_i \leq a_i$ for $i = 1, \dots, n$ then $\mathbf{a} \in \mathcal{A}$ implies $\mathbf{a}' \in \mathcal{A}$.

Definition 4 (Kernel)

A *kernel* is a maximal independent set of vertices of G , i.e., $A \subseteq V(G)$ is a kernel if

- (i) no two vertices of A are connected in G ,
- (ii) each $x \in V(G) \setminus A$ is connected to at least one $y \in A$.

Definition 5 (Graph Entropy: Geometric Definition)

Let G be a graph on vertex set $V(G) = \{1, \dots, n\}$ and let $P = (p_1, \dots, p_n)$ be a probability distribution on $V(G)$ (i.e., $p_1 + \dots + p_n = 1$ and $p_i \geq 0$ for all i). The entropy of G with respect to P is then defined as:

$$H(G, P) = \min_{\mathbf{a} \in VP(G)} \sum_{i=1}^n p_i \log_2 \frac{1}{a_i}.$$

Definition 6 (Graph Entropy: Information-Theoretic)

Let G be a graph and let P be a probability distribution on $V(G)$. The graph entropy of G with respect to P is defined by:

$$H(G, P) = \min I(X; Y),$$

where the minimum is taken over all pairs of random variables (X, Y) satisfying:

- ① X takes values in $V(G)$;
- ② Y takes values in the family of kernels of G ;
- ③ $\Pr(X \in Y) = 1$;
- ④ The marginal distribution of X is P .

Definition 7 (Co-normal Power)

Given a graph $G = (V, E)$, the t -th co-normal power G^t of G is defined by

$$V(G^t) = [V(G)]^t = \underbrace{V(G) \times \cdots \times V(G)}_{t \text{ times}},$$

and the edge set $E(G^t)$ is given by

$$E(G^t) = \left\{ \{\mathbf{x}, \mathbf{y}\} \mid \exists i \in \{1, \dots, t\} \text{ such that } \{x_i, y_i\} \in E(G) \right\},$$

where $\mathbf{x} = (x_1, \dots, x_t)$ and $\mathbf{y} = (y_1, \dots, y_t)$ are vertices in $V(G^t)$.

Definition 8 (Graph Entropy: Chromatic Limit)

Let $G = (V, E)$ be a graph and P a probability distribution on V . The *graph entropy* of G with respect to P is defined as

$$H(G, P) = \lim_{t \rightarrow \infty} \min_{\substack{U \subseteq V^t \\ P^t(U) > 1-\lambda}} \frac{1}{t} \log_2 \chi(G^t(U)),$$

for a fixed $0 < \lambda < 1$, where $P^t(U) = \sum_{\mathbf{x} \in U} P^t(\mathbf{x})$, for $\mathbf{x} = (x_1, \dots, x_t) \in V^t$ the product distribution is $P^t(\mathbf{x}) = \prod_{i=1}^t P(x_i)$, and $\chi(H)$ denotes the chromatic number of H .

Definition 9 (Cartesian Power)

Let $G = (V, E)$ be a graph and P a probability distribution on V . The cartesian power G_π^t of G has $V(G_\pi^t) = V(G^t)$ and

$$E(G_\pi^t) = \left\{ \{\mathbf{x}, \mathbf{y}\} \mid \forall i \in \{1, \dots, t\} \text{ either } \{x_i, y_i\} \in E(G) \text{ or } x_i = y_i \right\},$$

where $\mathbf{x}, \mathbf{y} \in V(G_\pi^t)$. Also, for $\mathbf{x} \in V(G_\pi^t)$, $P_\pi^t(\mathbf{x}) = \prod_{i=1}^t P(x_i) = P^t(\mathbf{x})$.

Remark: $\overline{G^t} = (\overline{G})_\pi^t$, and $\overline{G_\pi^t} = (\overline{G})^t$.

Definition 10 (P -typical Sequence)

Given a finite set X and a probability distribution P on it, a sequence $\mathbf{x} \in X^n$ is called P -typical if for each $y \in X$, the number of occurrences of y in $\mathbf{x}$, $N(y \mid \mathbf{x})$, satisfies

$$|N(y \mid \mathbf{x}) - nP(y)| \leq K\sqrt{nP(y)},$$

for a constant K .

Definition 11 (Fractional Chromatic Number)

Let $G = (V, E)$ be a graph, and let $S(G)$ denote the family of all independent sets of G . The *fractional chromatic number* of G , denoted by $\chi_f(G)$, is defined as:

$$\chi_f(G) = \min \sum_{S \in S(G)} w_S,$$

subject to

$$\sum_{\substack{S \in S(G) \\ v \in S}} w_S \geq 1 \quad (\forall v \in V), \quad w_S \geq 0 \quad (\forall S \in S(G)).$$

Theorem 12 (Equivalence of Graph Entropy Definitions)

Let $G = [V, E, P]$ be a finite probabilistic graph. Then, the following quantities are identical:

$$H_1(G, P) = \lim_{t \rightarrow \infty} \min_{\substack{U \subseteq V^t \\ P^t(U) > 1 - \varepsilon}} \frac{1}{t} \log \chi(G^t(U)),$$

$$H_2(G, P) = \min_{\substack{X \in Y \in \mathcal{S}(G) \\ P_X = P}} I(X; Y),$$

$$H_3(G, P) = \min_{\mathbf{a} \in VP(G)} - \sum_{i=1}^n p_i \log a_i,$$

where random variables X, Y take values in $V(G)$ with marginal P , and the set of kernels $\mathcal{S}(G)$, respectively, and $0 < \varepsilon < 1$ is fixed. That is,

$$H_1(G, P) = H_2(G, P) = H_3(G, P).$$

Lemma 13

Let $S(\cdot)$ denote the family of kernels. Then, $S(G^n) = S(G)^n$.

Lemma 14

The chromatic number $\chi(G)$ of an arbitrary graph equals the minimum number of kernels covering G .

Lemma 15

Let $T^n(P)$ denote the set of P -typical n -sequences, defined in Definition 10. Then:

(i) $\forall \lambda, \exists K$ such that for this K ,

$$P(\overline{T^n(\mathbf{P})}) < \lambda,$$

where

$$P(\overline{T^n(\mathbf{P})}) = \sum_{x \in \overline{T^n(\mathbf{P})}} P^n(x),$$

and $\overline{S}$ denotes the complement of the set S . (Every $K > \sqrt{|X|/\lambda}$ is suitable).

(ii) For every $x \in T^n(P)$,

$$2^{-(nH(P)+C\sqrt{n})} \leq P^n(x) \leq 2^{-(nH(P)-C\sqrt{n})}.$$

(iii) For every $n \in \mathbb{N}$,

$$2^{nH(P)-C\sqrt{n}} \leq |T^n(P)| \leq 2^{nH(P)+C\sqrt{n}}.$$

The constant C appearing in both (ii) and (iii) does not depend on n and P ; it depends only on $|X|$ and λ .

Proof of Lemma 15 (part 1)

Proof of (i): Under the binomial model, $N(y \mid \mathbf{x})$ has mean $\mu = n\mathbf{P}(y)$ and variance bound $\sigma^2 \leq n\mathbf{P}(y)$. Applying Chebyshev's inequality,

$$\mathbf{P}(|Z - \mu| > k) \leq \frac{\sigma^2}{k^2}$$

and setting $Z = N(y \mid \mathbf{x})$ with $k = K\sqrt{n\mathbf{P}(y)}$ yields:

$$\mathbf{P}\left(|N(y \mid \mathbf{x}) - n\mathbf{P}(y)| > K\sqrt{n\mathbf{P}(y)}\right) \leq \frac{\sigma^2}{K^2 n\mathbf{P}(y)} \leq \frac{1}{K^2}$$

Proof of Lemma 15 (part 2)

Proof of (ii): We find the total sequence probability $\mathbf{P}(\mathbf{x})$ by multiplying symbol probabilities based on how often each symbol appears, given by $N(y \mid \mathbf{x})$. Because these counts stay close to average for typical sequences, entropy lets us trap $\mathbf{P}(\mathbf{x})$ tightly:

$$2^{-(nH(\mathbf{P})+C\sqrt{n})} \leq \mathbf{P}(\mathbf{x}) \leq 2^{-(nH(\mathbf{P})-C\sqrt{n})}$$

Proof of (iii): By summing sequence probabilities whose total is bounded between $1 - \lambda$ and 1, we trap the number of typical sequences $|T^n(\mathbf{P})|$ tightly:

$$2^{nH(\mathbf{P})-C\sqrt{n}} \leq |T^n(\mathbf{P})| \leq 2^{nH(\mathbf{P})+C\sqrt{n}}$$

Lemma 16

Let $\mathbf{e} \in T^n(P_E)$ be a typical sequence of equivalence classes. The number of typical sequences contained within the clique $\mathbf{e}$ satisfies:

$$2^{nH(P|e)-C\sqrt{n}} \leq |T^n(P) \cap \mathbf{e}| \leq 2^{nH(P|e)+C\sqrt{n}},$$

where $C > 0$ is a numerical constant independent of n .

- **Sequence Decomposition:**

Typicality of the sequence $\iff$ conditional typicality of restricted subsequences.

$\implies$ Total size $|T^n(P) \cap \mathbf{e}| = \mathbf{Product}$ of conditionally typical set sizes.

- **Deviation Control:**

Since $\mathbf{e}$ is typical, class occurrence counts stay bounded near theoretical means.

$\implies$ Controls statistical deviations to establish the final bound.

Lemma 17

If $x \sim y \Leftrightarrow \{ \text{either } (x, y) \in E(X) \text{ or } x = y \}$ is an equivalence relation on the vertex set (i.e. G is the union of pairwise disjoint complete subgraphs), then

$$2^{nH(\mathbf{P}|e)-K\sqrt{n}} \leq \chi_\pi(n, \lambda) \leq 2^{nH(\mathbf{P}|e)+K\sqrt{n}}$$

where

$$H(\mathbf{P} \mid e) = \sum_{x \in X} P(x) \log_2 \frac{P(e(x))}{P(x)}$$

and $e(x)$ denotes the union of the elements equivalent to x . K is a numerical constant independent of n and $\mathbf{P}$ but dependent of $\|X\|$ and λ .

Proof of Lemma 17 (part1)

$$\textbf{Lower Bound:} \quad \chi_\pi(n, \lambda) \geq 2^{nH(\mathbf{P}|e) - K\sqrt{n}}$$

Let $S \subseteq X^n$ be the optimal subset with $P(S) \geq 1 - \lambda$ and $\chi(G^n[S]) = \chi_\pi(n, \lambda)$.

By Lemma 3(i), choosing $P(\overline{T^n(\mathbf{P})}) < \epsilon$ yields the typical intersection bound:

$$P(S \cap T^n(\mathbf{P})) \geq P(S) - P(\overline{T^n(\mathbf{P})}) \geq 1 - \lambda - \epsilon = \delta > 0$$

$$P(\mathbf{x}) \leq 2^{-nH(\mathbf{P}) + C_1\sqrt{n}} \quad \forall \mathbf{x} \in S \cap T^n(\mathbf{P})$$

$$\delta \leq P(S \cap T^n(\mathbf{P})) \leq |S \cap T^n(\mathbf{P})| \cdot 2^{-nH(\mathbf{P}) + C_1\sqrt{n}}$$

$$|S \cap T^n(\mathbf{P})| \geq \delta \cdot 2^{nH(\mathbf{P}) - C_1\sqrt{n}} \geq 2^{nH(\mathbf{P}) - C'_1\sqrt{n}}$$

$$C'_1 = C_1 + \log_2(\delta)$$

Proof of Lemma 17 (part2)

for any arbitrary independent set I we have:

$$|I \cap C(\mathbf{e})| \leq 1 \quad \text{for any independent set } I$$

$$|I \cap S \cap T^n(\mathbf{P})| \leq |T^n(\mathbf{P}_E)|$$

$$|I \cap S \cap T^n(\mathbf{P})| \leq 2^{nH(\mathbf{P}_E) + C_2\sqrt{n}}$$

Since $S \cap T^n(\mathbf{P}) \subseteq S$, the total chromatic number is bounded by the intersection:

$$\begin{aligned} \chi_\pi(n, \lambda) &\geq \frac{|S \cap T^n(\mathbf{P})|}{\max_I |I \cap S \cap T^n(\mathbf{P})|} \\ &\geq \frac{2^{nH(\mathbf{P}) - C'_1\sqrt{n}}}{2^{nH(\mathbf{P}_E) + C_2\sqrt{n}}} \\ &= 2^{nH(\mathbf{P}|e) - K\sqrt{n}} \quad \blacksquare \end{aligned}$$

Proof of Lemma 17 (part3)

$$\textbf{Upper Bound:} \quad \chi_{\pi}(n, \lambda) \leq 2^{nH(\mathbf{P}|e) + K\sqrt{n}}$$

$$\exists K > \sqrt{|X|/\lambda} \implies P(\overline{T^n(\mathbf{P})}) < \lambda$$

$$P(T^n(\mathbf{P})) \geq 1 - \lambda$$

$$\chi_{\pi}(n, \lambda) \leq \chi\left(G^n[T^n(\mathbf{P})]\right)$$

if a sequence is typical, its corresponding equivalence class sequence is also typical:

$$\mathbf{x} \in T^n(\mathbf{P}) \implies \mathbf{e} = e(\mathbf{x}) \in T^n(\mathbf{P}_E)$$

This ensures that all vertices in $T^n(\mathbf{P})$ belong exclusively to typical equivalence classes.

Proof of Lemma 17 (part4)

$G^n[T^n(\mathbf{P})]$ is a disjoint union of cliques $C(\mathbf{e})$

$$M_{\max} = \max_{\mathbf{e} \in T^n(\mathbf{P}_E)} |T^n(\mathbf{P}) \cap C(\mathbf{e})| \leq 2^{nH(\mathbf{P}|e) + C_3\sqrt{n}}$$

$$\chi(G^n[T^n(\mathbf{P})]) = M_{\max}$$

$$\chi_\pi(n, \lambda) \leq \chi(G^n[T^n(\mathbf{P})]) = M_{\max} \leq 2^{nH(\mathbf{P}|e) + K\sqrt{n}} \quad \blacksquare$$

Proposition 18

$$H_2(G, P) \leq \liminf_{n \rightarrow \infty} \min_{\substack{U \subseteq V^n \\ P^n(\bar{U}) > 1 - \lambda}} \frac{1}{n} \log_2 \chi(G^n(U))$$

Proof of Proposition 18 (Step 1)

Proof: To show that

$$H_2(G, P) \leq \liminf_{n \rightarrow \infty} \min_{\substack{U \subseteq V^n \\ P^n(U) > 1 - \lambda}} \frac{1}{n} \log_2 \chi(G^n(U)),$$

it suffices to show that for any $U \subseteq V^n$ satisfying $P^n(U) > 1 - \lambda$ and any arbitrary $\delta > 0$, for all sufficiently large n :

$$\chi(G^n(U)) \geq 2^{n(H_2(G, P) - \delta)}.$$

Proof of Proposition 18 (Step 2)

Using typical sequences (with $P(T^n(P)) \geq 1 - \varepsilon$), we bound $\chi(U \cap T^n(P))$ from below:

$$\begin{aligned} 1 - \varepsilon - \lambda &\leq P(U \cap T^n(P)) \\ &\leq \chi(U \cap T^n(P)) \cdot \max_{\mathbf{A} \in S^n(G)} P(\mathbf{A} \cap T^n(P)) \end{aligned}$$

Yielding the core inequality:

$$\chi(U \cap T^n(P)) \geq \frac{1 - \varepsilon - \lambda}{\max_{\mathbf{A} \in S^n(G)} P(\mathbf{A} \cap T^n(P))}$$

Proof of Proposition 18 (Step 3)

To upper bound the denominator, we factor $\max_{\mathbf{A} \in S^n(G)} P(\mathbf{A} \cap T^n(P))$ into the sequence probability and kernel cardinality:

$$\max_{\mathbf{A} \in S^n(G)} P(\mathbf{A} \cap T^n(P)) \leq \left(\max_{\mathbf{x} \in T^n(P)} P(\mathbf{x}) \right) \cdot \left(\max_{\mathbf{A} \in S^n(G)} |\mathbf{A} \cap T^n(P)| \right).$$

By the typical sequences lemma, the maximum sequence probability satisfies:

$$\max_{\mathbf{x} \in T^n(P)} P(\mathbf{x}) \leq 2^{-(nH(\mathbf{P}) - C\sqrt{n})}.$$

Thus, it remains to bound the maximum kernel size $\max_{\mathbf{A} \in S^n(G)} |\mathbf{A} \cap T^n(P)|$.

Proof of Proposition 18 (Step 4)

Next, to bound the maximum kernel size $\max_{\mathbf{A} \in S(G)^n} |\mathbf{A} \cap T^n(P)|$, let $\mathcal{Q}$ denote the set of admissible joint distributions $Q(x, A)$ with marginal $P(x)$ and $Q(x, A) = 0$ for $x \notin A$.

We aim to show that:

$$\max_{\mathbf{A} \in S(G)^n} |\mathbf{A} \cap T^n(P)| \leq (n+1)^{|V(G) \times S(G)|} \cdot 2^{n \cdot \max_{Q' \in \mathcal{Q}} H(Q'|a) + K_2 \sqrt{n}}.$$

Consider the joint sequence formed by aligning $\mathbf{x}$ and $\mathbf{A}$:

$$(\mathbf{x}, \mathbf{A}) = \begin{pmatrix} x_1 & x_2 & \dots & x_n \\ A_1 & A_2 & \dots & A_n \end{pmatrix}.$$

Proof of Proposition 18 (Step 5a)

For each $\mathbf{x}' \in \mathbf{A} \cap T^n(P)$, define the empirical joint distribution $Q_{\mathbf{x}'}$ by:

$$Q_{\mathbf{x}'}(y, B) = \frac{N(y, B \mid \mathbf{x}', \mathbf{A})}{N(y \mid \mathbf{x}')} \cdot P(y).$$

Thus, $(\mathbf{x}', \mathbf{A})$ is $Q_{\mathbf{x}'}$ -typical and $\mathbf{A}$ is typical under its marginal distribution. Since the marginal on $V(G)$ is P , we have $Q_{\mathbf{x}'} \in \mathcal{Q}$. Summing over these distributions yields:

$$\max_{\mathbf{A} \in S(G)^n} \left| \mathbf{A} \cap T^n(P) \right| \leq \sum_{Q_{\mathbf{x}'}: \mathbf{x}' \in \mathbf{A} \cap T^n(P)} \left| \{ \mathbf{x} \in \mathbf{A} \mid (\mathbf{x}, \mathbf{A}) \text{ is } Q_{\mathbf{x}'}\text{-typical} \} \right|.$$

Proof of Proposition 18 (Step 5b)

Since there are at most $(n+1)^{|V(G) \times S(G)|}$ distinct joint distributions, bounding the sum by the maximum term gives:

$$\max_{\mathbf{A} \in S(G)^n} \left| \mathbf{A} \cap T^n(P) \right| \leq (n+1)^{|V(G) \times S(G)|} \max_{Q' \in \mathcal{Q}} \left| \{ \mathbf{x} \in \mathbf{A} \mid (\mathbf{x}, \mathbf{A}) \text{ is } Q'\text{-typical} \} \right|.$$

To bound the Q' -typical pairs in $\mathbf{A}$, define the auxiliary graph Γ :

$$\begin{aligned} V(\Gamma) &= \left\{ (x, A) : x \in V(G), A \in S(G), x \in A \right\}, \\ E(\Gamma) &= \left\{ ((x_1, A_1), (x_2, A_2)) : A_1 \neq A_2 \right\}. \end{aligned}$$

Proof of Proposition 18 (Step 5c)

Here, non-adjacency ($A = B$) induces an equivalence relation a , making each equivalence class a clique in $\bar{\Gamma}$.

Applying Lemma 16 to $\bar{\Gamma}$ along sequence $\mathbf{A}$ bounds the number of Q' -typical sequence pairs by $2^{nH(Q'|a)+K_2\sqrt{n}}$, yielding the final bound:

$$\max_{\mathbf{A} \in S(G)^n} \left| \mathbf{A} \cap T^n(P) \right| \leq (n+1)^{|V(G) \times S(G)|} \cdot 2^{n \cdot \max_{Q' \in \mathcal{Q}} H(Q'|a) + K_2\sqrt{n}}.$$

Proof of Proposition 18 (Step 6a)

Substituting the bound from the previous step yields:

$$\chi(U \cap T^n(P)) \geq (1-\varepsilon-\lambda) \cdot 2^{n \left(H(P) - \max_{Q' \in \mathcal{Q}} H(Q'|a) \right) - K\sqrt{n} - |V(G) \times S(G)| \log_2(n+1)}.$$

Since P is the marginal distribution over $V(G)$, we have:

$$H(P) - \max_{Q' \in \mathcal{Q}} H(Q'|a) = \min_{Q' \in \mathcal{Q}} [H(P) - H(Q'|a)] = \min_{X \in Y \subseteq S(G)} I(X; Y) = H_2(G, P).$$

Since $\chi(G^n(U)) \geq \chi(U \cap T^n(P))$, substituting $H_2(G, P)$ gives:

$$\chi(G^n(U)) \geq 2^{n \left(H_2(G, P) + \frac{\log_2(1-\varepsilon-\lambda)}{n} - \frac{K}{\sqrt{n}} - \frac{|V(G) \times S(G)| \log_2(n+1)}{n} \right)}.$$

Proof of Proposition 18 (Step 6b)

Since all non-entropy terms inside the exponent vanish as $n \rightarrow \infty$, for any $\delta > 0$ and all sufficiently large n :

$$\chi(G^n(U)) \geq 2^{n(H_2(G,P)-\delta)}.$$

Finally, taking $\frac{1}{n} \log_2(\cdot)$ on both sides and taking the limit inferior as $n \rightarrow \infty$ concludes the proof:

$$H_2(G, P) \leq \liminf_{n \rightarrow \infty} \min_{\substack{U \subseteq V^n \\ P^n(U) > 1-\lambda}} \frac{1}{n} \log_2 \chi(G^n(U)). \quad \blacksquare$$

Proposition 19

$$\limsup_{n \rightarrow \infty} \min_{\substack{U \subseteq V^n \\ P^n(\bar{U}) > 1 - \lambda}} \frac{1}{n} \log_2 \chi(G^n(U)) \leq H_2(G, P)$$

Proof of Proposition 19 (Part 1)

Proof: Let $S(G)$ denote the family of kernels of G . We fix a joint distribution $P_{X,Y}$ on $V(G) \times S(G)$ supported on $\{(x, Y) : x \in Y\}$ that minimizes mutual information, so that $I(X; Y) = H_2(G, P)$.

This induces a marginal distribution $P_Y(Y) = \sum_{x \in Y} P_{X,Y}(x, Y)$, which we extend to $S(G)^n$ via the product distribution

$$P_{Y^n}(\mathbf{Y}) = \prod_{i=1}^n P_Y(Y_i), \quad \text{for } \mathbf{Y} = (Y_1, \dots, Y_n) \in S(G)^n.$$

To cover a high-probability subset of $V(G)^n$, we select K sets $\mathbf{Y}_1, \dots, \mathbf{Y}_K \in S(G)^n$ independently and identically distributed (i.i.d.) according to P_{Y^n} , with joint distribution $P_K[\mathbf{Y}_1, \dots, \mathbf{Y}_K] = \prod_{j=1}^K P_{Y^n}(\mathbf{Y}_j)$.

Also, we define $U[\mathbf{Y}_1, \dots, \mathbf{Y}_K] = \bigcup_{j=1}^K \mathbf{Y}_j \subseteq V(G)^n$ to be the set of sequences covered by $\mathbf{Y}_1, \dots, \mathbf{Y}_K$.

Proof of Proposition 19 (Part 2)

Take $K = \lfloor 2^{n(H_2(G,P)+\delta)} \rfloor$ where $\delta > 0$ is arbitrary, and suppose there exists $\mathbf{Y}_1^*, \dots, \mathbf{Y}_K^*$ such that

$$P^n(U^c[\mathbf{Y}_1^*, \dots, \mathbf{Y}_K^*]) < \lambda, \quad \text{equally} \quad P^n(U[\mathbf{Y}_1^*, \dots, \mathbf{Y}_K^*]) \geq 1 - \lambda.$$

Then, defining $T_n = U[\mathbf{Y}_1^*, \dots, \mathbf{Y}_K^*]$, by Lemma 14, and 13, we have

$$\chi(G^n(T_n)) \leq K = \lfloor 2^{n(H_2(G,P)+\delta)} \rfloor.$$

Consequentially:

$$\min_{\substack{U \subseteq V^n \\ P^n(U) > 1-\lambda}} \frac{1}{n} \log_2 \chi(G^n(U)) \leq \frac{1}{n} \log_2 \chi(G^n(T_n)) \leq H_2(G, P) + \delta$$

$$\implies \limsup_{n \rightarrow \infty} \min_{\substack{U \subseteq V^n \\ P^n(U) > 1-\lambda}} \frac{1}{n} \log_2 \chi(G^n(U)) \leq H_2(G, P) + \delta$$

which completes the proof.

Proof of Proposition 19 (Part 3)

Building on this, it is enough to prove the existence of such a $\mathbf{Y}_1^*, \dots, \mathbf{Y}_K^*$. For this matter, we prove for a randomly selected $\mathbf{Y}_1, \dots, \mathbf{Y}_K$, and sufficiently large n ,

$$\mathbb{E}_{P_K} [P^n (U^c[\mathbf{Y}_1, \dots, \mathbf{Y}_K])] < \lambda.$$

We have:

$$\mathbb{E}_{P_K} [P^n (U^c[\mathbf{Y}_1, \dots, \mathbf{Y}_K])] = \sum_{\mathbf{x} \in V(G)^n} P^n(\mathbf{x}) \Pr_{P_K}(\mathbf{x} \text{ is not covered})$$

where $\Pr_{P_K}(\mathbf{x} \text{ is not covered})$ is the probability that the sequence $\mathbf{x} \in V(G)^n$ is not contained in any of $\mathbf{Y}_1, \dots, \mathbf{Y}_K$.

Proof of Proposition 19 (Part 4)

We partition $V(G)^n$ into typical sequences $T^n(P)$ and non-typical sequences $\overline{T^n(P)}$. So that:

$$\mathbb{E}_{P_K} [P^n(U^c[\mathbf{Y}_1, \dots, \mathbf{Y}_K])] = \sum_{\mathbf{x} \in T^n(P)} P^n(\mathbf{x}) \Pr_{P_K}(\mathbf{x} \text{ is not covered}) + \sum_{\mathbf{x} \in \overline{T^n(P)}} P^n(\mathbf{x}) \Pr_{P_K}(\mathbf{x} \text{ is not covered}) \quad (1)$$

Proof of Proposition 19 (Part 5)

For the second part, it is clear that

$$\sum_{\mathbf{x} \in \overline{T^n(P)}} P^n(\mathbf{x}) \Pr_{P_K}(\mathbf{x} \text{ is not covered}) < \sum_{\mathbf{x} \in \overline{T^n(P)}} P^n(\mathbf{x}) = P^n(\overline{T^n(P)}).$$

By Lemma 15 (i), for any $\lambda > 0$, there exists a constant C_1 such that

$$P^n(\overline{T^n(P)}) < \frac{\lambda}{2},$$

which concludes:

$$\sum_{\mathbf{x} \in \overline{T^n(P)}} P^n(\mathbf{x}) \Pr_{P_K}(\mathbf{x} \text{ is not covered}) < \frac{\lambda}{2}. \quad (2)$$

Proof of Proposition 19 (Part 6)

For the first part, we have

$$\Pr_{P_K}(\mathbf{x} \text{ is not covered}) = \left[1 - \sum_{\mathbf{y} \in \mathbf{Y} \in S(G)^n} P_{Y^n}(\mathbf{Y}) \right]^K.$$

Using the inequality $(1 - t)^K \leq \exp_2(-Kt)$, we get

$$\left[1 - \sum_{\mathbf{y} \in \mathbf{Y} \in S(G)^n} P_{Y^n}(\mathbf{Y}) \right]^K \leq \exp_2 \left(-K \sum_{\mathbf{y} \in \mathbf{Y} \in S(G)^n} P_{Y^n}(\mathbf{Y}) \right). \quad (3)$$

Proof of Proposition 19 (Part 7)

To bound the number $N(\mathbf{x})$ of typical covering sequences $\mathbf{Y} \in S(G)^n$ for a typical sequence $\mathbf{x} \in T^n(P)$, we define the auxiliary graph $\tilde{\Gamma}$ on $V(G) \times S(G)$ with equivalence relation $(x_1, A_1) \sim (x_2, A_2) \iff x_1 = x_2$, such that

$$\begin{aligned} V(\tilde{\Gamma}) &= \{(x, A) : x \in V(G), A \in S(G), x \in A\}, \quad \text{and} \\ E(\tilde{\Gamma}) &= \{((x_1, A_1), (x_2, A_2)) : x_1 = x_2\}. \end{aligned}$$

The equivalence classes $e(x) = \{(x, A) : x \in A\}$ partition $V(\tilde{\Gamma})$ into disjoint cliques.

Extending componentwise to $\tilde{\Gamma}^n$, the product equivalence class

$$e^n(\mathbf{x}) = e(x_1) \times \cdots \times e(x_n)$$

forms a disjoint clique in $\tilde{\Gamma}^n$, representing all pairs $(\mathbf{x}, \mathbf{Y})$ with $x_i \in Y_i$.

Since $P(e(x_i)) = P(x_i)$, the induced distribution $P(e^n)$ coincides with P on $V(G)^n$, implying that $e^n(\mathbf{x})$ is P -typical whenever $\mathbf{x}$ is P -typical. Thus, $\tilde{\Gamma}^n$ satisfies the conditions of Lemma 16.

Proof of Proposition 19 (Part 8)

Note that for each $(\mathbf{x}, \mathbf{Y}) \in T^n(P_{X,Y})$ we have $\mathbf{Y} \in T^n(P_{Y^n})$.

Now, by Lemma 16, the number $N(\mathbf{x})$ of the covering sequences $\mathbf{Y} \in S(G)^n$ satisfies the following bound:

$$2^{nH(P_{X,Y}|e) - C_2\sqrt{n}} \leq |T^n(P_{X,Y}) \cap e^n(\mathbf{x})| \leq N(\mathbf{x}),$$

where $C_2 > 0$ is a numerical constant independent of n .

Furthermore, by Lemma 15, each such $P_{X,Y}$ -typical sequence $\mathbf{Y}$ satisfies

$$P_{Y^n}(\mathbf{Y}) \geq 2^{-nH(P_Y) - C_3\sqrt{n}},$$

where $C_3 > 0$.

Proof of Proposition 19 (Part 9)

Summing $P_{Y^n}(\mathbf{Y})$ over all such typical $\mathbf{Y}$ yields

$$\begin{aligned}\sum_{\mathbf{x} \in \mathbf{Y} \in S(G)^n} P_{Y^n}(\mathbf{Y}) &\geq 2^{nH(P_{X,Y}|e) - C_2\sqrt{n}} \cdot 2^{-nH(P_Y) - C_3\sqrt{n}} \\ &= 2^{-nH_2(G,P) - C_4\sqrt{n}}.\end{aligned}$$

Using (3), and $K = \lfloor 2^{n(H_2(G,P) + \delta)} \rfloor$ yields:

$$\begin{aligned}\left[1 - \sum_{\mathbf{x} \in \mathbf{Y} \in S(G)^n} P_{Y^n}(\mathbf{Y}) \right]^K &\leq \exp_2 \left(-2^{n(H_2(G,P) + \delta)} \cdot 2^{-nH_2(G,P) - C_4\sqrt{n}} \right) \\ &= \exp_2 \left(-2^{n\delta - C_4\sqrt{n}} \right).\end{aligned}$$

Proof of Proposition 19 (Part 10)

For any fixed $\delta > 0$, $n\delta - C_4\sqrt{n} \rightarrow \infty$ as $n \rightarrow \infty$. Consequently,

$$\exp_2 \left(-2^{n\delta - C_4\sqrt{n}} \right) \rightarrow 0$$

. Thus, for sufficiently large n ,

$$\left[1 - \sum_{\mathbf{x} \in \mathbf{Y} \in S(G)^n} P_{Y^n}(\mathbf{Y}) \right]^K < \frac{\lambda}{2}, \quad \forall \mathbf{x} \in T^n(P).$$

Proof of Proposition 19 (Part 11)

Now, we have

$$\sum_{\mathbf{x} \in T^n(P)} P^n(\mathbf{x}) \Pr_{P_K}(\mathbf{x} \text{ is not covered}) < \frac{\lambda}{2} \sum_{\mathbf{x} \in T^n(P)} P^n(\mathbf{x}) < \frac{\lambda}{2}.$$

Combining this with (1) and (2),

$$\mathbb{E}_{P_K} [P^n(U^c[\mathbf{Y}_1, \dots, \mathbf{Y}_K])] < \frac{\lambda}{2} + \frac{\lambda}{2} = \lambda,$$

for sufficiently large n , and we are done. ■

Proposition 20

$$H_2(G, P) \geq H_3(G, P)$$

Proof of Proposition 20 (Part 1)

Proof: Let $V(G) = \{1, \dots, n\}$ and let $\mathcal{S}(G)$ denote the set of maximal independent sets.

Let (X, Y) attain $\min I(X; Y)$ with $P_X = P = (p_1, \dots, p_n)$. We define:

$$Q(J) = \Pr(Y = J), \quad J \in \mathcal{S}(G)$$

$$R(J \mid i) = \begin{cases} \Pr(Y = J \mid X = i) & \text{if } i \in J \\ 0 & \text{if } i \notin J \end{cases}$$

By total probability representation:

$$Q(J) = \sum_{i=1}^n \Pr(Y = J, X = i) = \sum_{i=1}^n p_i R(J \mid i)$$

Proof of Proposition 20 (Part 2)

The mutual information can be expressed as:

$$\begin{aligned} I(X; Y) &= - \sum_{J \in \mathcal{S}(G)} Q(J) \log_2 Q(J) + \sum_{i=1}^n p_i \sum_{J \in \mathcal{S}(G)} R(J | i) \log_2 R(J | i) \\ &= - \sum_{i=1}^n p_i \sum_{\substack{J \in \mathcal{S}(G) \\ i \in J}} R(J | i) \log_2 \frac{Q(J)}{R(J | i)}. \end{aligned}$$

Applying Jensen's inequality to the inner sum yields:

$$\sum_{\substack{J \in \mathcal{S}(G) \\ i \in J}} R(J | i) \log_2 \frac{Q(J)}{R(J | i)} \leq \log_2 \left(\sum_{\substack{J \in \mathcal{S}(G) \\ i \in J}} Q(J) \right).$$

Proof of Proposition 20 (Part 3)

Consequently, we have:

$$I(X; Y) \geq - \sum_{i=1}^n p_i \log_2 \left(\sum_{\substack{J \in \mathcal{S}(G) \\ i \in J}} Q(J) \right).$$

Define $\mathbf{a} = (a_1, \dots, a_n)$ by:

$$a_i = \sum_{\substack{J \in \mathcal{S}(G) \\ i \in J}} Q(J) \implies \mathbf{a} = \sum_{J \in \mathcal{S}(G)} Q(J) \mathbf{1}_J.$$

Since Q is a probability distribution on $\mathcal{S}(G)$, by Definition 2:

$$\mathbf{a} \in VP(G).$$

Proof of Proposition 20 (Part 4)

It follows directly that:

$$I(X; Y) \geq - \sum_{i=1}^n p_i \log_2 a_i \geq \min_{\mathbf{a} \in VP(G)} \left\{ - \sum_{i=1}^n p_i \log_2 a_i \right\}.$$

Since (X, Y) was chosen to attain the minimum in $H_2(G, P)$, we conclude:

$$\min_{\substack{X \in Y \in \mathcal{S}(G) \\ P_X = P}} I(X; Y) \geq \min_{\mathbf{a} \in VP(G)} \left\{ - \sum_{i=1}^n p_i \log_2 a_i \right\}.$$

Hence, $H_2(G, P) \geq H_3(G, P)$. ■

Proposition 21

$$H_2(G, P) \leq H_3(G, P)$$

Proof of Proposition 21 (Part 1)

Proof: Let $\mathbf{a} \in \text{VP}(G)$ be the minimizing vector in $H_3(G, P)$. Expressing $\mathbf{a}$ as a convex combination of maximal independent sets:

$$a_i = \sum_{J \in S(G)} Q'(J), \quad \text{where} \quad \sum_{J \in S(G)} Q'(J) = 1.$$

We define the conditional probability distribution $R'(J \mid i)$ and the marginal distribution $Q^*(J)$ on $S(G)$ as:

$$R'(J \mid i) = \begin{cases} \frac{Q'(J)}{a_i}, & i \in J, \\ 0, & \text{otherwise,} \end{cases} \quad \text{and} \quad Q^*(J) = \sum_{i=1}^n p_i R'(J \mid i).$$

Proof of Proposition 21 (Part 2)

Considering the pair of random variables (X, Y) with $\text{dist}(X) = P$, $\text{dist}(Y) = Q^*$, and $\text{dist}(Y | X) = R'$. Following the same proof as in the previous part, we obtain

$$I(X; Y) = - \sum_{i=1}^n p_i \sum_{\substack{J \in \mathcal{S}(G) \\ i \in J}} R'(J | i) \log_2 \frac{Q^*(J)}{R'(J | i)}.$$

Therefore, the mutual information satisfies:

$$\min_{X \in \mathcal{Y} \in \mathcal{S}(G), P_X = P} I(X; Y) \leq - \sum_{i=1}^n p_i \sum_{J \in \mathcal{S}(G)} R'(J | i) \log_2 \frac{Q^*(J)}{R'(J | i)}. \quad (1)$$

Proof of Proposition 21 (Part 3)

Using Jensen's inequality and the concavity of the logarithm, we have

$$\sum_{J \in S(G)} Q^*(J) \log_2 \frac{Q'(J)}{Q^*(J)} \leq 0$$

which implies

$$-\sum_{i,J} p_i R'(J | i) \log_2 Q^*(J) \leq -\sum_{i,J} p_i R'(J | i) \log_2 Q'(J).$$

Combining this with (1), we obtain

$$\min_{X \in Y \in S(G), P_X = P} I(X; Y) \leq -\sum_{i,J} p_i R'(J | i) \log_2 \frac{Q'(J)}{R'(J | i)}.$$

Proof of Proposition 21 (Part 4)

Substituting $\frac{Q'(J)}{R'(J|i)} = a_i$ yields:

$$-\sum_{i,J} p_i R'(J|i) \log_2 \frac{Q'(J)}{R'(J|i)} = -\sum_{i=1}^n p_i \log_2 a_i,$$

Since a was chosen to attain the minimum in $H_3(G, P)$, we conclude that

$$\min_{X \in Y \in S(G), P_X = P} I(X; Y) \leq \min_{\mathbf{a} \in VP(G)} -\sum_{i=1}^n p_i \log_2 a_i,$$

which means $H_2(G, P) \leq H_3(G, P)$. $\blacksquare$

Proof of Theorem 12

Proof: By Proposition 20 and 21, we have $H_2(G, P) = H_3(G, P)$. Denoting $\chi(n, \lambda) := \min_{U \subseteq V^n, P^n(U) > 1-\lambda} \chi(G^n(U))$, combining Propositions 19 and 18 yields:

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log_2 \chi(n, \lambda) \leq H_2(G, P) \leq \liminf_{n \rightarrow \infty} \frac{1}{n} \log_2 \chi(n, \lambda).$$

Since $\liminf \leq \limsup$ always holds, the limit exists and equals $H_2(G, P)$, which implies $H_1(G, P) = H_2(G, P)$.

Consequently, we conclude:

$$H_1(G, P) = H_2(G, P) = H_3(G, P). \quad \blacksquare$$

Theorem 22

$$\max_Q H(G, Q) = \log_2 \chi_f(G),$$

where $\chi_f(G)$ is the fractional chromatic number of graph G .

Proof of Theorem (Part 1: Upper Bound)

Let $P = (p_1, \dots, p_n)$ attain the maximum graph entropy. By the backward closure property of the packing polytope:

$$\left(\frac{1}{\chi_f(G)}, \dots, \frac{1}{\chi_f(G)} \right) \in VP(G).$$

This directly implies:

$$H(G, P) \leq \sum_{i=1}^n p_i \log_2 \chi_f(G) = \log_2 \chi_f(G) \sum_{i=1}^n p_i = \log_2 \chi_f(G).$$

Therefore, we have the upper bound:

$$\max_Q H(G, Q) = H(G, P) \leq \log_2 \chi_f(G).$$

Proof of Theorem (Part 2: Lower Bound)

By Lemma 24, G contains an induced subgraph G' with $\chi_f(G') = \chi_f(G)$ such that for any $w \in VP(G')$ with $w_i \geq \frac{1}{\chi_f}$ for all i , equality holds for every i .

Since

$$\left(\frac{1}{\chi_f}, \dots, \frac{1}{\chi_f}\right)_{1 \times |V(G')|} \in VP(G'),$$

by Lemma 25, there exists a probability distribution P' on $V(G')$ such that:

$$H(G', P') = \sum_{i=1}^{|V(G')|} p'_i \log_2 \left(\frac{1}{\frac{1}{\chi_f(G)}} \right) = \log_2 \chi_f(G).$$

Extending P' to P on $V(G)$ by setting $p_i = 0$ for all $i \in V(G) \setminus V(G')$ yields:

$$\log_2 \chi_f(G) = H(G, P) \leq \max_Q H(G, Q). \quad \blacksquare$$

Lemma 23

Let $G = (V, E)$ be a graph, and let $\{\lambda_{I_i}\}_{i=1}^t$ be an optimal weight assignment to its independent sets $\{I_i\}_{i=1}^t$ attaining the fractional chromatic number $\chi_f(G)$. For each vertex $v \in V(G)$, define its weight as:

$$w(v) = \sum_{i=1}^t \lambda_{I_i}.$$

If there exists a vertex $v \in V(G)$ such that $w(v) > 1$, then removing v does not change the fractional chromatic number of G , i.e.,

$$\chi_f(G \setminus \{v\}) = \chi_f(G).$$

Proof of Lemma 23 (Part 1)

Proof: First, we show that $\chi_f(G \setminus \{v\}) \leq \chi_f(G)$.

Suppose I is an independent set of G . Then $I \setminus \{v\}$ is an independent set of both G and $G \setminus \{v\}$, and these sets form the complete family of independent sets of $G \setminus \{v\}$.

We assign the weights $\{\lambda'_{I_i \setminus \{v\}}\}_{i=1}^t$ to the independent sets of $G \setminus \{v\}$ by

$$\lambda'_{I \setminus \{v\}} = \begin{cases} \lambda_I + \lambda_{I \setminus \{v\}}, & I \cup \{v\} \text{ independent in } G, \\ \lambda_{I \setminus \{v\}}, & I \cup \{v\} \text{ not independent in } G. \end{cases}$$

Since the weight of every vertex other than v remains unchanged,

$$\chi_f(G \setminus \{v\}) \leq \sum_{J \in \{I_i \setminus \{v\}\}} \lambda'_J = \sum_{i=1}^t \lambda_{I_i} = \chi_f(G).$$

Proof of Lemma 23 (Part 2)

Next, by contradiction, assume that

$$\chi_f(G \setminus \{v\}) < \sum_{i=1}^t \lambda_{I_i}.$$

Then there exists a weighting $\{\gamma_{I_i \setminus \{v\}}\}_{i=1}^t$ such that

$$\sum_{J \in \{I_i \setminus \{v\}\}} \gamma_J < \sum_{i=1}^t \lambda_{I_i}.$$

Define a new weighting for each I_i , denoted by λ''_{I_i} ,

$$\lambda''_{I_i} := \begin{cases} \frac{1}{w(v)} \lambda_{I_i}, & v \in I_i, \\ \frac{1}{w(v)} \lambda_{I_i} + \left(1 - \frac{1}{w(v)}\right) \gamma_{I_i \setminus \{v\}}, & v \notin I_i. \end{cases}$$

Proof of Lemma 23 (Part 3)

For this weighting, we have $\lambda''_{I_i} \geq 0$ for every i , $\sum_{u \in I_i} \lambda''_{I_i} \geq 1$ for each vertex u , and

$$\sum_{i=1}^t \lambda''_{I_i} < \sum_{i=1}^t \lambda_{I_i} = \chi_f(G)$$

Therefore, by the definition of the fractional chromatic number:

$$\chi_f(G) \neq \sum_{i=1}^t \lambda_{I_i},$$

which contradicts our initial assumption.

Thus,

$$\chi_f(G \setminus \{v\}) = \chi_f(G). \quad \blacksquare$$

Lemma 24

G have an induced subgraph G' such that $\chi_f(G') = \chi_f(G) =: \chi_f$, and for any $w \in VP(G')$ satisfying $\forall i : w_i \geq \frac{1}{\chi_f}$, we have equality for every i .

Proof of Lemma 24 (Part 1)

Proof: We iteratively remove every vertex whose weight is strictly greater than 1 under an optimal weighting for χ_f , which does not change χ_f using Lemma 23. Hence, we obtain an induced subgraph G' such that $\chi_f(G') = \chi_f(G) =: \chi_f$, and every optimal weighting of G' , w , satisfies

$$w(v) = 1 \quad \text{for every } v \in V(G').$$

We aim to prove that G' is indeed the desired induced subgraph in the lemma.

Proof of Lemma 24 (Part 2)

Suppose, for contradiction, that there exists $y \in VP(G')$ such that

$$\forall i, \quad y_i \geq \frac{1}{\chi_f}, \quad \text{and} \quad y_j > \frac{1}{\chi_f}$$

for some j .

Since $y \in VP(G')$, we can write

$$y = \sum_{i=1}^{t'} \alpha_i \mathbf{1}_{I_i}, \quad \alpha_i \geq 0, \quad \sum_{i=1}^{t'} \alpha_i = 1.$$

Now coefficients $\{\alpha_i \chi_f\}_{i=1}^{t'}$ of independent sets.

This forms an optimal weighting, yielding χ_f .

Proof of Lemma 24 (Part 3)

According to the initial assumption, the weight of every vertex must be equal to 1. However, for vertex v_j , we have:

$$w(v_j) = \sum_{v_j \in I_i} \alpha_i \chi_f = \chi_f \sum_{v_j \in I_i} \alpha_i = \chi_f y_j > \chi_f \left(\frac{1}{\chi_f} \right) = 1,$$

which is a contradiction.

Consequently, no such vector y exists, and we are done. $\blacksquare$

Lemma 25

Suppose $a \in VP(G)$ is a maximal vector in $VP(G)$, i.e., there exists no vector $a' \in VP(G) \setminus \{a\}$ such that $a' \geq a$ coordinatewise. Then, there exists a probability distribution P on $V(G)$ such that the value of $H(G, P)$ is attained at a .

Proof of Lemma 25 (Part 1)

Since $a \in A$ is a maximal element, $a \in \partial^+ A$.

By a result of the supporting hyperplane theorem, there exists $b = (b_1, \dots, b_k) \in \mathbb{R}_+^k$, $b \neq 0$, such that

$$b^T x \leq 1 \quad \forall x \in A, \quad b^T a = \sum_{i=1}^k b_i a_i = 1.$$

Define

$$p_i = a_i \cdot b_i,$$

that defines a probability distribution P on $V(G)$. Thus,

$$H(G, P) = H_{VP(G)}(p) = \min_{x \in VP(G)} \left(- \sum_{i=1}^k p_i \log_2 x_i \right).$$

Proof of Lemma 25 (Part 2)

Let $x = (x_1, \dots, x_k)$ be an arbitrary vector in $VP(G)$. Applying Jensen's inequality to the strictly concave logarithm function, we obtain:

$$-\sum_{i=1}^k p_i \log_2 a_i \leq -\sum_{i=1}^k p_i \log_2 x_i$$

for all $x \in VP(G)$.

Therefore, the convex objective function $f(x) = -\sum_{i=1}^k p_i \log_2 x_i$ achieves its minimum over $VP(G)$ at $x = a$, and the value of $H(G, P)$ is attained by a . ■

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